Activity: ZIP models

Weekend drinking

In the first part of this class activity, we work with survey data from 77 college students on a dry campus (i.e., alcohol is prohibited) in the US. The survey asks students "How many alcoholic drinks did you consume last weekend?" The data includes the following variables:

- `drinks`: the number of drinks the student reports consuming
- `sex`: an indicator for whether the student identifies as male (1 = male)
- `OffCampus`: an indicator for whether the student lives off campus (1 = off campus, 0 = on campus)
- `FirstYear`: an indicator for whether the student is a first-year student (1 = first year, 0 = not first year)

The model

To account for excess zeros in the data, we fit a *zero-inflated Poisson* (ZIP) model. Here is the probability function for our model:

$$P(Y_i = y|\mathbf{x}_i) = \begin{cases} e^{-\lambda_i}(1-p_i) + p_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-p_i) & y > 0 \end{cases}$$

where

$$\log\left(\frac{p_i}{1-p_i}\right) = \mathbf{x}_i^T\gamma$$
$$\log(\lambda_i) = \mathbf{x}_i^T\beta$$

Model output

```
m1 <- zeroinfl(drinks ~ sex + FirstYear + OffCampus,
               dist = "poisson", data = wdrinks)

summary(m1)
```

```
Call:
zeroinfl(formula = drinks ~ sex + FirstYear + OffCampus, data = wdrinks,
    dist = "poisson")

Pearson residuals:
    Min      1Q  Median      3Q     Max
-1.4561 -0.8381 -0.4783  0.5688  5.2796

Count model coefficients (poisson with log link):
              Estimate Std. Error z value Pr(>|z|)
(Intercept)     0.8010     0.1620   4.945 7.60e-07 ***
sexm            0.9835     0.1889   5.205 1.94e-07 ***
FirstYearTRUE  -0.1619     0.3651  -0.444   0.6574
OffCampusTRUE   0.3724     0.2135   1.744   0.0811 .

Zero-inflation model coefficients (binomial with logit link):
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -0.39619    0.39752  -0.997    0.319
sexm          -0.07079    0.58846  -0.120    0.904
FirstYearTRUE  0.89197    0.65878   1.354    0.176
OffCampusTRUE -1.69137    1.47761  -1.145    0.252
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 14
Log-likelihood: -139.6 on 8 Df
```

Questions

1. What is the estimated probability that a female first-year student who lives on campus never drinks?

2. What is the estimated average number of drinks for a second-year male student who lives off campus and sometimes drinks?

3. What is the estimated probability that a male first-year student who lives off campus had at least one drink last weekend?